



Words or Pictures: A comparison of written and pictorial explanations of physical and chemical equilibria

Sevil Akaygun & Loretta L. Jones

To cite this article: Sevil Akaygun & Loretta L. Jones (2014) Words or Pictures: A comparison of written and pictorial explanations of physical and chemical equilibria, International Journal of Science Education, 36:5, 783-807, DOI: [10.1080/09500693.2013.828361](https://doi.org/10.1080/09500693.2013.828361)

To link to this article: <https://doi.org/10.1080/09500693.2013.828361>



Published online: 12 Aug 2013.



Submit your article to this journal [↗](#)



Article views: 1496



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 7 View citing articles [↗](#)

Words or Pictures: A comparison of written and pictorial explanations of physical and chemical equilibria

Sevil Akaygun^{a*} and Loretta L. Jones^b

^aSecondary School Science & Mathematics Education, Bogazici University, Istanbul, Turkey; ^bChemistry and Biochemistry, University of Northern Colorado, 501 20th Street, Greeley, CO 80639, USA

The features of a concept or principle an individual chooses to highlight in an explanation or description may be related to the medium of communication used. Different aspects of understanding can be revealed through words and through drawings. This two-part exploratory study examined the differences between explanations of physical and chemical equilibria generated by means of words or pictures. Participants included both instructors and students, who were randomly assigned to provide either written explanations or drawings of physical or chemical equilibrium at the macroscopic and particulate levels. For both studies, analyses revealed that significantly different features appeared in the written and pictorial explanations of equilibrium. The written responses focused more on processes such as the dynamic nature of equilibrium, whereas pictorial representations highlighted structural aspects of equilibrium, such as the spatial arrangement of molecules. Regardless of the level of chemistry knowledge, people conveyed the same type of information via the use of different representations and visual tools.

Keywords: *Explanations; Drawings; Representations; Equilibrium; Chemistry*

Introduction

Tversky (2001) has shown that both written language and visualizations are cognitive tools and that, in order to understand how people think, it is important to understand both their writings and their pictorial representations. Modern scientific explanations often include pictures as well as text. However, little is known about which aspects of scientific concepts are best represented in text and which in pictures (Cheng & Gilbert, 2009). University instructors and high school teachers use both kinds of

*Corresponding author: Secondary School Science & Mathematics Education, Bogazici University, Istanbul, Turkey. Email: sevil.akaygun@boun.edu.tr

representations in their classes. Chemistry students may use both kinds of explanations when they answer conceptual questions and some research studies assess student understanding by having them draw pictures of their mental models as well as explain them in words (Kelly & Jones, 2007; Niaz, 1995; Nyachwaya et al., 2011; Sanger, 2000).

Knowing which features of a chemistry concept are more easily represented in written and pictorial communications may allow for more accurate measurement of conceptual understanding. The purpose of this study was to examine the differences between written and pictorial explanations generated for the concept of equilibrium by participants with a broad range of backgrounds.

Theoretical Background

Mental Models

Mental models are the ‘small-scale models’ of reality created by people as a result of their perception, imagination, experience, and interaction; or the comprehension of discourse as visual images or abstract representations of situations that cannot be visualized (Coll & Treagust, 2003; Craik, 1943; Greca & Moreira, 2000). Mental models can be represented by both textual and pictorial tools. Paivio (1990) classifies physical representations, such as drawings, maps, and diagrams, as *picture-like* and language, mathematics and symbolic logic as *language-like*. Even though they both have variations from concrete to abstract in terms of structure and function, picture-like representations are highly concrete and language-like ones more abstract.

Models and modeling are especially important in chemistry because chemical phenomena can be represented by concrete, visual, mathematical, and/or verbal modes of representation. Therefore, the role of models is central to learning chemistry and includes coming to know already produced models, appreciating the role of models in the communication of chemical knowledge, and creating and testing models (Justi & Gilbert, 2002). Learner-generated mental models of different chemical phenomena have been investigated in a number of studies that reveal the level of conceptual understanding as well as the way learners visualize the phenomena (Coll & Treagust, 2003; Harrison & Treagust, 2002; Ingham & Gilbert, 1991).

Various methods have been used to elicit mental models. Although some researchers (Greca & Moreira, 2000) suggest that it is not possible to access an individual’s mental models, most researchers accept the notion of ‘expressed model’ (Gilbert, 1997), which is a mental model that an individual makes public by some form of communication. For identifying complex mental representations, open-ended research methods may be more revealing than techniques such as multiple-choice questions (Bodner, Gardner, & Briggs, 2005; Lesh, Hover, Hole, Kelly, & Post, 2000; Tversky et al., 2006). Dynamic equilibrium is a scientific topic that includes complex mental representations such as molecular structure and dynamics. Therefore, an open-ended questionnaire was used in this study.

Recently, researchers have used qualitative tools such as drawings (Davidowitz, Chittleborough, & Murray, 2010) and open-ended assessments (Kern, Wood, Roehrig, & Nyachwaya, 2010) to understand students' conceptualizations of chemical phenomena. Nyachwaya et al. (2011) reviewed existing diagnostic tools used to uncover students' alternative conceptions and claimed that the existing tools are not effective at providing in-depth understanding of students' mental models. They proposed drawing tasks as alternative diagnostic tools to uncover students' understandings and misconceptions.

For chemical educators, mental models are important constructs to identify because they convey how people understand and represent chemical phenomena. Noting how different types of cognitive tools such as text or drawings are used to emphasize different aspects of the same phenomenon may give insight to educators and instructional designers. Effective learning environments may then be created with text and drawing tools that correspond to preferably used modes of communication.

Pictorial and Written Explanations

Information can be presented either in words or pictures or both. Words or verbal form refers to printed or spoken text, whereas pictures refer to static graphics, illustrations, photos, maps, or dynamic graphics (Mayer, 2001, p. 2). Pictorial representations, such as cave paintings, rock engravings, wood carvings, pictures on pottery, and designs in weavings, were created long before written language was used and represent meaning more directly than language. The use of pictures developed into written languages to describe concepts difficult to depict, such as abstract entities, causality, and quantification (Tversky, 2001).

Tversky argues that, like language, external images or diagrams reflect how people think and visualize things (Tversky, 2005). These depictions simplify and schematize information. Therefore, in order to design effective instruction, it is important to identify which types of features need to be included in the pictorial representations for a specific topic and when text is required. In order to properly assess student understanding, it is important to know when to ask for a written explanation and when to ask for a drawing. A systematic comparison of the pictorial and verbal tools used in student explanations of chemical concepts and principles has been carried out for the first time in this study in order to inform these types of decisions.

Science has a discourse of mixed or multi-modal forms of representations (Ainsworth, 1999). Instructors and students are expected to know and use these representations and to be able to translate from one to another. Prain and Tytler (2012) suggest that the tools used in explanations are not just simple representations of conceptual knowledge, but are also means to understand how knowledge is developed and communicated in the scientific community. For this reason, it is important to consider the specific affordances that particular tools may offer when people make claims about scientific concepts or processes because they may constrain the way people convey information. For instance, representing a process or a change by a

drawing on paper may be limited to the availability of physical space, conventions used, and details to be provided (Prain & Tytler, 2012). Written representations that describe a scientific phenomenon in words convey the sequence of causal mechanisms in a linear manner; on the other hand, they may suffer from the limitations of portraying the spatial aspects in words, whereas diagrams and models (static and dynamic) provide good media to convey these spatial attributes (Gobert, 2005).

Amongst the various tools used to represent chemical concepts, visual images have prime importance because chemistry involves the visualization of structures, behaviors, and processes. Gobert (2005) describes visualizations as external, internal, or spatial skills. External visualizations are the ones used for learning, internal visualizations refer to the mental models people have, and visualizations as spatial skills refer to the ability to manipulate or transform spatial patterns. These skills are not mutually exclusive and are particularly important in chemistry, because most of the phenomena involve specific action or interaction of the particles. External visualizations are extensively used in teaching chemistry, and recently a number of research studies (Ehrlén, 2009; Kelly & Jones, 2007; Kozma & Russell, 1997; Prain & Tytler, 2012) have asked experts and students to generate visual explanations in the form of drawings or diagrams to display their internal visualizations or 'expressed models' (Gilbert, 1997). However, in previous research it was observed that when students were asked to generate explanations by using both text and images, they appeared to use text to communicate different ideas than they communicated with their drawings (Ardac & Akaygun, 2005; Kelly & Jones, 2008). Therefore, collecting only written explanations or drawings may give incomplete data on understanding.

This study was designed to investigate differences between the features emphasized in written and pictorial explanations for equilibrium, and to identify the specific affordances of these two types of communication. In other words, this study is an attempt to better understand the strengths and limitations of these two types of communication by comparing the specific features used to explain the nature of equilibrium. By being aware of the specific affordances of each type of representation, instructors and instructional designers may better decide *when* and *how* to make use of each type of representation for specific purposes.

Understanding the Observable and Particulate Levels of Matter

Gilbert and Treagust (2009) argue that understanding the ideas of chemistry involves mental engagement with chemical phenomena through the use of different types of representations. Learning chemistry involves conceptualizing chemical phenomena at three levels: the macroscopic level of the observable and visible; the submicroscopic level of the molecular, atomic, and kinetic; and the symbolic level of symbols, equations, and mathematics (Johnstone, 1993). The symbolic level is the one most often emphasized in chemistry classes and macroscopic level phenomena are the ones easily observed. Yet understanding chemistry at the macroscopic and symbolic levels does not always imply an understanding of the concepts at the submicroscopic level. Research by Margel, Eylon, and Scherz (2008) showed that students have

difficulty in understanding the particulate nature of matter (PNM) and has revealed related misconceptions (Ben-Zvi, Eylon, & Silberstein, 1986; Coll & Treagust, 2003; Harrison & Treagust, 2002; Nakhleh, 1992).

Nurrenbern and Pickering (1987) expressed concern that students who appear to understand molecular-level concepts may have only surface understandings. They showed that learners who could correctly answer textual questions failed to answer questions on the same concepts when pictures had to be interpreted in order to answer the question. Similar results were reported by Smith and Metz (1996). One concern with these findings is that students may have difficulty understanding novel pictorial representations of molecular phenomena (Cheng & Gilbert, 2009). Although Lekhavat and Jones (2009) did not find an effect of pictorial novelty in a study of learning from multimedia, the ease of interpretation of molecular visualizations can vary widely (Jones, Jordan, & Stillings, 2005; Noh & Scharmann, 1997).

Williamson (2008) reviewed theory-based research on the PNM and its impact on learning. The author claimed that students' misconceptions or difficulty in conceptualizing the PNM can arise when students apply explanations for the observable level to the particulate level, or when they are unable to visualize, diagram or depict the behavior of particles (p. 72).

Understanding Physical and Chemical Equilibria

Dynamic equilibrium has been rated as one of the most difficult chemistry topics for students to understand due to its conceptual complexity (Bergquist & Heikkinen, 1990; Camacho & Good, 1989; Chiu, Chou, & Liu, 2002; Gussarsky & Gorodetsky, 1988, 1990; Hackling & Garnett, 1985; Tyson, Treagust, & Bucat, 1999; Wheeler & Kass, 1978). Some of the misconceptions about chemical equilibrium that have been reported in the literature are: no discrimination between reactions that go to completion and reversible reactions (Banerjee, 1995; Wheeler & Kass, 1978); having equal concentrations at equilibrium (Tyson et al., 1999); holding a static rather than dynamic understanding of the nature of the system at chemical equilibrium (Chiu et al., 2002; Gussarsky & Gorodetsky, 1990); and confusion regarding amount and concentration, specifically oscillation behavior (Bergquist & Heikkinen, 1990). In a review of the literature on chemical equilibrium, Van Driel and Gräber (2002) highlighted the difficulties students experience in understanding the dynamic nature of equilibrium, and suggested that more qualitative data on student models of equilibrium are needed.

The literature on physical equilibrium is less extensive than that on chemical equilibrium, but some studies have reported misconceptions about physical change and physical equilibrium. Osborne and Cosgrove (1983) reported that one of the common misconceptions amongst 8- to 17-year-old students was the breaking of water into its component hydrogen and oxygen molecules or atoms upon boiling. Haidar and Abraham (1991) identified students' misconceptions about physical and chemical changes, such as involvement of chemical reactions in physical processes of dissolution, using macroscopic reasoning for molecular explanations, and focusing

on a single factor in a process. Ebenezer and Gaskell (1995) reported specific misconceptions held by students about solutions, such as sugar changing from solid sugar to liquid sugar on dissolution.

Various methods have been used to identify conceptual difficulties that students have in understanding dynamic equilibrium, including a test of misconceptions (Wheeler & Kass, 1978), two-tiered tests (Tyson et al., 1999; Voska & Heikkinen, 2000), word association techniques (Gussarsky & Gorodetsky, 1988), phenomenography (Ebenezer & Gaskell, 1995), and a cognitive apprenticeship approach (Chiu et al., 2002). Regardless of the methodology used, each of these studies found that students have difficulty understanding the concept of dynamic equilibrium.

In summary, students have difficulty in understanding dynamic physical and chemical equilibria, particularly concepts involving dynamic change at the particulate level. One of the reasons students experience these conceptual barriers could be the difficulty of visualizing the dynamic nature of the equilibrium. For this reason, dynamic equilibrium (physical and chemical) was the topic selected as the focus of this study.

Purpose of the Study

Although having students draw pictures and write explanations to assess their understanding is a well-established practice, there is a need for research that investigates the nature of textual and pictorial assessments (Cheng & Gilbert, 2009; Williamson, 2008). Dynamic equilibrium, both physical and chemical, is an important topic in chemistry that requires conceptual understanding of the PNM and dynamic molecular processes. This study was part of a larger investigation of mental models of equilibrium. It is original because it investigates how written and pictorial explanations differ and categorizes the features appearing in representations of mental models of a specific chemistry topic. This article reports on the differences in written and pictorial explanations and illustrates how the characteristics of pictorial and verbal explanations can be analyzed to determine mental models. The research question was:

In which aspects do pictorial and written explanations of physical and chemical equilibria at the macroscopic and particulate levels differ?

Method

A cognitive science approach outlined by Tversky et al. (2006) was chosen as the basis for the methods used in this study. In the first part of this approach, the mental representations of people for a given domain and the visual devices they use to convey them are revealed by having them write explanations illustrated with drawings. This study differs in that participants were asked either to write or to draw explanations, in order to compare the kinds of information each type of communication provides. The study had two parts: the physical equilibrium study (PES) and the chemical equilibrium study (CES). The procedure was the same in each case, but the chemistry

content and participants differed. Data were collected by open-ended questionnaires, quantified by coding, and confirmed by inter-rater check. For triangulation of data, three instructors and three students were interviewed to explain what they had drawn or written. The transcribed interviews were compared with the drawn and written responses as a credibility check. In addition, the inter-rater reliability check confirmed the minimization of the subjective interpretation of the responses, particularly the pictorial ones. For the inter-rater reliability, the coded responses were checked by two researchers and percentage agreement was 96%.

Participants

Participants were drawn from a variety of backgrounds to ensure a wide spectrum of responses. Therefore, participants of the studies included both instructors and students, all of whom were volunteers. Instructors included university chemistry ($N = 21$) or biology ($N = 2$) instructors, high school chemistry teachers ($N = 14$) and chemistry ($N = 39$) or biology ($N = 2$) graduate students. All the university instructors and graduate students had experience teaching general chemistry, and the high school teachers were currently teaching chemistry. One hundred students were drawn from a first-year college chemistry course and 65 from a high school chemistry course in the western USA. Specifically, 47 college students who were taking first semester chemistry, and 47 high school students who were taking chemistry in their second year in high school participated in the PES. The average age of college students was 19 and that of high school students was 16. Similarly, in the CES, 53 college students who were taking second semester general chemistry and 18 high school students who were taking chemistry in the Spring semester of their second year of chemistry participated. The average ages of college and high school students were 20 and 16, respectively. All students had previously studied the concept of dynamic equilibrium. The questions asked in the open-ended questionnaire were selected from instances with which they were expected to be familiar based on the chemistry instruction they had received. However, in their tests or homework, the students had encountered primarily numerical or qualitative equilibrium problems. They had not viewed animations of dynamic equilibrium, nor had they been trained in drawing or explaining their ideas, so completing the questionnaire might have been a challenge for them.

Informed consent was obtained from all the participants and from the parents of the high school students. As described below, participants in each category were randomly assigned to communicate using either written or pictorial representations. Table 1 shows the number of participants in each category.

Instruments

The physical equilibrium questionnaire (PEQ) and chemical equilibrium questionnaire (CEQ) were prepared by the researchers and validated by a group of chemistry experts. The PEQ questions were designed to evaluate conceptual understanding of

Table 1. Participants in the PES and CES

		Instructors			Students	
		INS	HST	GRS	FYC	HSS
PES	Written ($N = 70$)	10	6	8	21	25
	Pictorial ($N = 69$)	4	4	13	26	22
CES	Written ($N = 55$)	5	2	10	28	10
	Pictorial ($N = 49$)	4	2	10	25	8

Note: INS: University instructors; HST: high school teachers; GRS: graduate students; FYC: first-year college students; HSS: high school students.

liquid–vapor and dissolving–recrystallization (solubility) equilibria at the macroscopic and particulate levels in two forms: written and pictorial. In the PEQ, there were five questions on liquid–vapor equilibrium and three questions on solubility equilibrium. The CEQ contained a total of four questions about the dimerization reaction of nitrogen dioxide at the macroscopic and particulate levels before and at equilibrium, and 10 minutes after equilibrium had been established, again in written and pictorial forms. Participants were randomly assigned to answer the questions by providing either written explanations or drawings in 30 minutes (Table 1). All the participants answered the questions by using either a pen or pencil; no specific drawing instruments (e.g. colored pencils) were provided to the participants,

The PEQ and CEQ were administered to the instructors at individually scheduled times in a separate room. They were administered to first-year college students at the beginning of a laboratory session in the chemistry laboratory and to high school students during the first half of their chemistry class. Participants were also asked to complete a demographic questionnaire that asked about gender, ethnicity, and chemistry background. Instructors were asked about their teaching experience and students were asked about their expected chemistry grade. Figure 1 shows sample questions from the PEQ and CEQ.

Data Analysis

Mental models of the participants were elicited from the pictorial or written explanations they provided in their responses to the PEQ and CEQ. First, all responses were recorded, compared, and listed. In this process, the responses of all participants were first coded by open-coding, then categorized based on the features emphasized as well as by a rubric (see Table S1 in the supplementary materials). The rubric was developed by first listing all the key features that appeared in each response; similar categories were then combined. The numerical values in the rubric refer to different categories, not to a score; thus, the total or raw scores were not calculated. It should be emphasized that the goal of this study was not to examine how correct an explanation was but rather what features were used to communicate understanding. The coding scheme was designed to determine how mental models derived from drawings

- (a) Let's say you are heating some amount of water in two different containers by using the same heater. The first container is an open flask and the second one is a closed (sealed) flask. Water boils at 100°C and freezes at 0°C. Assume that the sealed flask remains intact while it is being heated.
(**Molecular Level is the level of atoms, ions and molecules**)



Sealed flask
before heating

Pictorial Form of PEQ:

Draw *the process(es)* occurring at the *macroscopic* and *particulate* levels at the surface of water that has been heated to 90 °C in both *open* and *closed (sealed)* flasks. That is, what would you see if you could see the molecules?

Written Form of PEQ:

Explain in your own words *the process(es)* occurring at the *macroscopic* and *particulate* levels at the surface of water that has been heated to 90 °C in *open* and *closed (sealed)* flasks. That is, what would you see if you could see the molecules?

- (b) Iodine, I₂, is a very slightly water soluble purple molecular solid that forms light tan solutions. Iodine crystals are added to the water in the container and stirred until no more can dissolve and some crystals remain at the bottom of the container. The solution is said to be *saturated* with iodine.
(**Molecular Level is the level of atoms, ions, and molecules**)

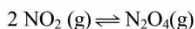
Pictorial Form of PEQ:

- 1) Draw what you would see with your own eyes in the saturated solution.
- 2) Draw what is happening at the molecular level in the saturated solution, showing any processes that may be occurring.

Written Form of PEQ:

- 1) Describe in your own words *what you would see with your own eyes* in the saturated solution.
- 2) Describe in your own words what is happening in the saturated solution at the *molecular level*, explaining any processes that may be occurring.

- (c) Let's say you add some brown NO₂ gas to an evacuated spherical flask and seal it. The molecules of NO₂ begin to join together in pairs to form colorless gaseous N₂O₄ molecules until the reaction reaches equilibrium at a certain constant temperature. At equilibrium there are twice as many N₂O₄ molecules as NO₂ molecules. Answer the following questions on the sheets of paper provided.
(**Molecular Level is the level of atoms, ions and molecules**)



Pictorial Form of CEQ:

- 1) Draw *what you would observe with your own eyes* in the sealed flask at four points in time as the reaction proceeds. That is, draw the appearance of the flask at (a) the initial state before reaction begins; (b) after some reaction has begun but prior to the equilibrium; (c) at equilibrium; and (d) 10 minutes later.
- 2) Draw *the process(es)* occurring at the *molecular level* in the sealed flask at four points in time as the reaction proceeds. That is, draw the molecules in the flask at (a) the initial state before reaction begins; (b) after some reaction has begun but prior to the equilibrium; (c) at equilibrium; and (d) 10 minutes later.

Written Form of CEQ:

- 1) Explain in detail *what you would observe with your own eyes* in the sealed flask at four points in time as the reaction proceeds. That is, describe the appearance of the flask at (a) the initial state before reaction begins; (b) after some reaction has begun but prior to the equilibrium; (c) at equilibrium; and (d) 10 minutes later.
- 2) Explain in your own words *the process(es)* occurring at the *molecular level* in the sealed flask at four points in time as the reaction proceeds. That is, describe what you would see if you could see the molecules in the flask at (a) the initial state before reaction begins; (b) after some reaction has begun but prior to the equilibrium, (c) at equilibrium and (d) 10 minutes later.

Figure 1. Pictorial and written forms of a PEQ question on (a) liquid–vapor equilibrium and (b) solubility equilibrium. (c) Pictorial and written forms of a CEQ question on chemical equilibrium

differ from mental models derived from written explanations. Therefore, for the purposes of this study all responses were coded and categorized without regard to the correctness of the answer. An example of the coding of a macroscopic level pictorial response for liquid–vapor equilibrium is given in Figure 2.

The same rubric (see Table S1) was used to code written and pictorial responses. Figure 3 shows an example of the coding of a particulate level written response for solubility equilibrium.

When some text, in the form of labels, was included in the drawings (Figure 2), the text passages were treated as a part of the explanations, so coded accordingly. In other words, text was considered as an element added for the clarification of the drawing.

In the analysis of the liquid–vapor equilibrium section of the PEQ, 16 features were identified for explanations of the macroscopic level and 20 features for explanations of the particulate level; for the solubility equilibrium section of the PEQ, 10 features were identified for the macroscopic level, and 19 features for the particulate level. In the analysis of the CEQ, 15 macroscopic level features and 17 particulate-level features were identified. Macroscopic level features also included *physical structures* which were observable objects such as flask, beaker, solid, or solution. Next, features common to both written and pictorial responses to questions on the macroscopic and particulate levels were coded as described in Figures 2 and 3. Then, 20% of all the papers were coded by a second researcher who was trained in the use of the rubric (Table S1). Inter-rater reliability that was obtained by calculating the percentage agreement of coding was 94% for the PES and 96% for the CES. After reaching inter-rater agreement by finding the percentage agreement, the coded responses were analyzed with χ^2 , assuring that all statistical restrictions of the χ^2 analysis were met.

Categories determined from pictorial and written responses were compared by χ^2 analysis. The rubric in Table S1 (supplementary materials) shows the list of features determined as a result of coding the responses of the participants. Then, the percentage of occurrence of each feature was calculated and similar categories were collapsed and



Flask: 1 (representation of a flask)
Macroscopic Feature: 4 (bubble and steam)
Gas: 1 (representation of gas)
Liquid: 1 (representation of liquid)
Surface: 1 (representation of surface)
Molecules: 0 (no representation of molecules)
Structure: 0 (no representation of structure)
Motion: 0 (no representation of motion)
Evaporation: 1 (representation of evaporation)
Condensation: 0 (no representation of condensation)
Process: 1 (representation of a process)
Dynamic nature: 0 (no implication of dynamic nature)
Pressure: 0 (no indication of pressure)
Temperature: 0 (no indication of temperature)
Rate: 0 (no indication of rate)
Misconception: 0 (no specific misconception)

Figure 2. An example of the coding of a macroscopic level pictorial response for liquid–vapor equilibrium

2. The iodine crystals are being broken up by the water molecules.

Flask: 0 (no representation of a flask)
 Bottom of the flask: 0 (no indication of bottom of the flask)
 Solid: 1 (representation of solid/crystals)
 Solution: 0 (no representation of solution)
 Surface: 0 (no representation of surface)
 Color: 0 (no representation of color)
 Molecules: 1 (representation of molecules)
 Structure: 2 (representation of structure in words, e.g. iodine, water)
 $\text{H}_2\text{O} - \text{I}_2$ Structures: 0 (no indication of H_2O - I_2 positioning around)
 Intermolecular interaction: 1 (indication of H_2O & I_2 intermolecular interaction)
 Saturation: 0 (no indication of saturation)
 Dissolving: 1 (indication of dissolving)
 Recrystallization: 0 (no indication of recrystallization)
 Process: 1 (indication of a process)
 Dynamic nature: 0 (no indication of dynamic nature)
 Equilibrium: 0 (no indication of equilibrium)
 Motion: 0 (no representation of motion)
 Rate: 0 (no indication of rate)
 Misconception: 0 (no specific misconception)

Figure 3. An example of the coding of a particulate level written response for solubility equilibrium

refined for the χ^2 analysis. For features that were found to be significantly different, percentage occurrences were calculated. For instance, for liquid–vapor equilibrium, at the macroscopic level, the frequency at which the flask appeared in responses was significantly different for the two groups: 88% (61 out of 69) of the pictorial responses but only 50% (35 out of 70) of the written responses included the flask.

Results and Discussion

This study focused on a limited area of chemistry and the results may not be transferable to other areas. In addition, in both parts of the study, subjects answered questions about the macroscopic level first; this may have affected their responses for the particulate level. Eleven students who were assigned to provide pictorial explanations omitted questions about the particulate level because they thought that they did not know how to draw the answers or they provided written explanations instead. Consequently, some data were missed. However, these limitations did not affect the findings of the study, which combined data from a total of 243 participants with a wide variety of backgrounds.

All the instructors had experience teaching equilibrium in introductory chemistry courses, either currently or in the past. No significant difference ($p > .05$) in features appearing in the responses was found between the answers of the different types of instructors in either study. Therefore, they were combined into one group of instructors. Similarly, the responses provided by college and high school students in both studies were not significantly different ($p > .05$); hence, they were combined into one student group. Participant demographics showed that for the PES 47% of the

instructors and 39% of the students were male. The average teaching experience of instructors was 9 years; specifically, college or university instructors had taught an average of 15 years, high school teachers 9 years, and graduate students 2 years. For the CES, 43% of the instructors and 38% of the students were male. Twenty-five percent of the instructors were teaching college-level chemistry, 14% of the instructors were high school teachers, and 61% of the instructors were graduate students in the field of chemistry or biology.

Significantly different features ($p < .05$) for written and pictorial explanations at the macroscopic and particulate levels for all the equilibrium systems were determined (Table 2).

Comparison of Written and Pictorial Descriptions of the Macroscopic Level

Macroscopic features that appeared more frequently in pictorial responses. Pictorial responses to questions on the macroscopic level included *physical structures* significantly more often ($p < .05$) than the written explanations in all three equilibrium cases. Specifically, flask, liquid phase, and surface for the liquid–vapor equilibrium; flask, bottom of the flask, solid, and solution for the solubility equilibrium; and flask, gas, and homogeneity for the chemical equilibrium appeared significantly more often in the pictorial responses.

Macroscopic features that appeared more often in written responses. Process features were mentioned more frequently in written explanations than in pictorial representations. Specifically, evaporation and process for the liquid–vapor equilibrium; color, name/formula, saturation, and representation of no-change for the solubility equilibrium; and name/formula, process, motion, and equilibrium for the chemical equilibrium were mentioned significantly more often in text than in pictures. Sample responses are given in Figure 4, which shows the explanations given by three students of what processes would be occurring at the macroscopic level for the three equilibrium systems. Specifically, in Figure 4(a), the drawing (on the left) shows only one type of process, evaporation, shown by the curved lines above the liquid, whereas the written explanation (on the right) refers to both evaporation and condensation by describing the motion of molecules as going ‘from liquid to gas then back into liquid’. The processes represented by pictorial and written explanations for solubility equilibria in Figure 4(b) differed in a similar fashion. The drawing (on the left) only shows only dissolving (the curved lines in the liquid), whereas the written explanation (on the right) mentions ‘the change in the shape of the crystal’ as not only an indication of dissolving but also an observable feature of a process. In the pictorial and written responses for chemical equilibrium, shown in Figure 4(c), similar descriptions of processes, such as *change in color*, were provided in the pictorial and written explanations. However, note that the participant asked to draw a response resorted to the use of words to describe the changes.

Table 2. Significantly different ($p < .05$) features of equilibrium systems in written and pictorial responses

	Macroscopic features	Pictorial (%)	Written (%)	Particulate features	Pictorial (%)	Written (%)
Liquid/vapor equilibrium study (PES)	Flask*	88	50	Macro features*	52	21
	Liquid phase*	93	43	Liquid phase*	71	30
	Surface*	93	30	Surface*	85	14
				Molecules	94	81
	Evaporation	12	26	Name/formula*	55	6
	Process*	29	64	Hydrogen bond*	18	3
				Pressure*	3	20
				Temperature	5	16
				Change as T increases*	23	36
				Rate*	15	50
				Motion	35	54
				Misconceptions*	5	20
				Process	42	61
Solubility equilibrium study (PES)	Flask*	95	37	Flask*	55	3
	Bottom of flask*	85	63	Bottom of flask*	52	18
	Solid	86	67	Solution*	83	61
	Solution*	97	81	Surface	22	9
				Molecules*	87	67
	Color	55	73	Name/formula*	58	35
	Name/formula	19	40	I ₂ -H ₂ O structure	55	36
	Saturation*	3	22			
	No change*	2	19	Saturation*	7	30
				Dissolving*	23	47
				Intermolecular attraction	18	36
				Process*	33	68
				Motion*	2	15
				Misconceptions	18	36
Chemical equilibrium study (CES)	Flask*	96	71	Flask*	70	26
	Gas*	80	53	Gas*	81	59
	Homogeneity*	88	31	Homogeneity*	52	9
				Name/formula	83	61
	Name/formula*	12	42	Equilibrium ratio*	49	18
	Process*	4	55	Misconceptions*		
	Motion*	2	18	– completion	11	20
	Equilibrium*	12	47	– proceeding	25	0
	Misconception*	54	69			
				Color change*	2	19
				Process*	20	82
				Dynamic nature*	12	40
				Motion*	8	64

* $p < .01$.

(a) 2.)



The water level would vary very little if at all because it is a closed system. The water molecules would go from liquid to gas then back into liquid because of pressure in the flask.

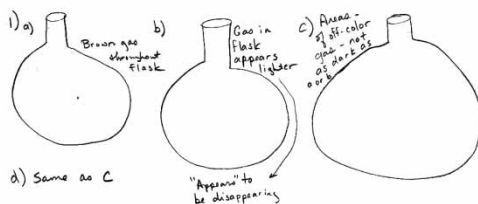
(b)



Full

I would see solid purple crystals of iodine at the bottom of the container. If I watched long enough (or observed it periodically) I would notice that the shape of the crystal(s) may change, but its total volume/size would not.

(c)



a) Flask contents are uniformly brown
b) Reaction begins and color begins to fade
c) At equilibrium, the color would be diminished
d) 10 minutes later, once equilibrium is established, no observable change.

Figure 4. Sample pictorial and written responses showing *process* for the macroscopic level explanations of (a) liquid–vapor equilibrium (in the drawing evaporation is indicated by curved lines above the liquid), (b) solubility equilibrium (in the drawing curved lines indicate crystals dissolving), and (c) chemical equilibrium; the participant instructed to draw found it necessary to use text to indicate process

Tversky (2001) suggested that pictures are effective cognitive tools when used as concrete representations and have more direct meanings than written language. On the other hand, languages can more effectively communicate abstract concepts, causality, and quantification, because it is difficult to depict these concepts pictorially. The findings of this study are found to be consistent with these ideas because more concrete or *physical structures* such as flask, solid, or surface appeared more often in pictorial representations; whereas more abstract concepts, such as evaporation, process, and the motion of molecules, were more likely to appear in written explanations. Even in responses to questions about the macroscopic (observable) level, most of the features emphasized significantly more often in the written responses were unobservable characteristics such as saturation and equilibrium. Similarly, the molecular formulae of the substances, a symbolic feature, appeared significantly more often in written explanations for the macroscopic level. In other words, in the written explanations of the macroscopic level, participants were more likely to provide the names or

chemical formulae of the particles than respondents who created drawings. In the drawings, formulae were used to label the locations of particles. An example of written macroscopic level explanation of a saturated solution is given below:

You'd see solid I_2 in the bottom of a flask containing a solution.

Comparison of Written and Pictorial Descriptions of the Particulate Level

Particulate-level features represented more often in pictorial responses. For items asking about the particulate level, *physical* or *macroscopic structures* such as the liquid surface or container appeared significantly more often ($p < .01$) in pictorial representations of all three equilibrium cases. Although these features are not necessary to describe the behavior of molecules at the particulate level they do provide readers or viewers with a setting in which to imagine the processes taking place, which may aid visualization.

In addition to macroscopic structures, *particulate structures* were also represented significantly more often in pictorial than written particulate-level explanations. For instance, as shown in Figure 5, molecules, formulae, and equilibrium ratio (for chemical equilibrium) were represented significantly more often ($p < .01$) in pictorial responses than in written responses. The responses in Figure 5 were representative attempts to explain (a) hydrogen bonding in water, (b) a saturated solution of iodine in water, and (c) four stages in the chemical equilibrium between NO_2 and N_2O_4 .

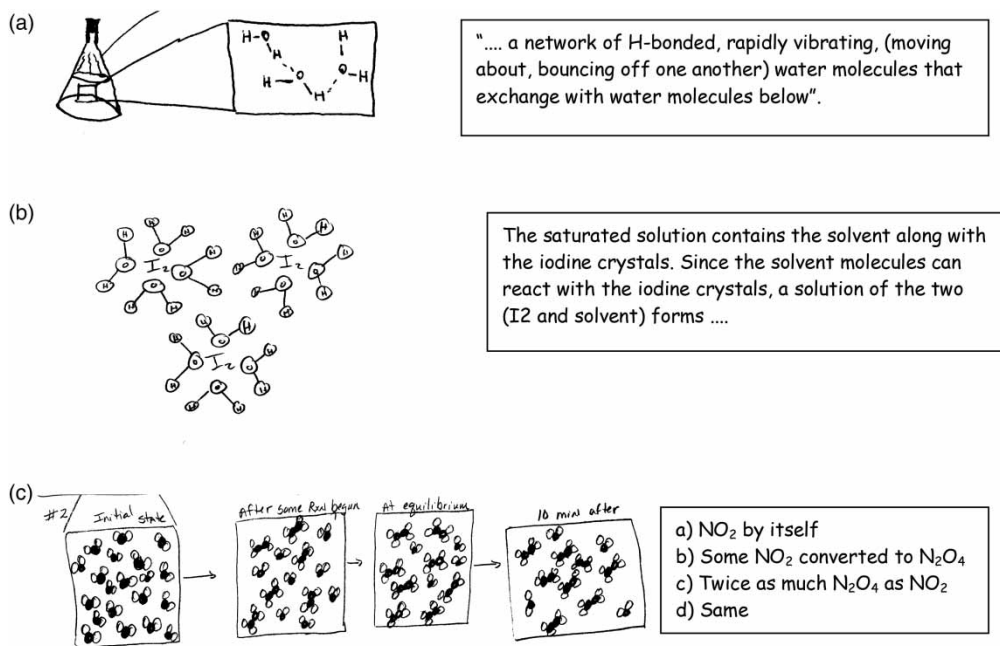


Figure 5. Sample pictorial and written responses describing *particulate structures* for the particulate-level explanations of (a) liquid–vapor equilibrium, (b) solubility equilibrium, and (c) chemical equilibrium

Specific features describing processes were coded from the written responses, but not from the pictorial ones, unless an attempt to show a process was observed. For instance, in Figure 5(a), the pictorial response (on the left) shows only the structural drawing of hydrogen bonding, whereas the written response (on the right) includes words such as *vibrating*, *bouncing*, and *exchanging*, which describe motion and interaction. Similarly, Figure 5(b), the pictorial response (on the left) only shows a saturated solution where I_2 molecules are surrounded by water molecules, representing the structure, whereas the text (on the right) explains the processes in terms of 'reaction' between water and iodine molecules. For the chemical equilibrium, the drawings (on the left) show the snapshots at specific time intervals, whereas the written explanation (on the right) describes the process by pointing out the reaction with the word 'conversion'.

Structures such as hydrogen bonds appeared significantly more often in drawings than in text explanations. One reason for this finding could be that these types of structures are easily shown in drawings but hard to explain in words. In addition, representing particulate-level structures such as hydrogen bonding or H_2O-I_2 interactions requires showing spatial relations amongst the particles; hence, they appeared more frequently in the pictorial representations. On the other hand, it may be that the writers simply did not think about these interactions when writing their explanations, while those generating drawings naturally included the interactions as they drew the molecules.

When coding data from particulate-level drawings, spatial orientations were considered both as a characteristic representing only the structure, and as spatial proximity conveying interactions. Specifically, for the solubility equilibrium study, some drawings only represented the structure of water and iodine as iodine was dispersed in water, as shown in Figure 6, whereas some drawings showed both the structures of the substances and their interaction due to spatial proximity as seen from Figure 5(b). In

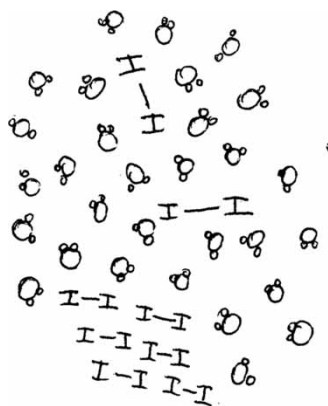


Figure 6. A sample pictorial response for the *particulate-level representation* of solubility equilibrium conveying spatial orientations where only the structure is shown. Compare this drawing to that in Figure 5(b), where the structure, orientation, and interaction of the substance were shown

other words, the drawing in Figure 6 was coded as ' I_2-H_2O interaction = 1' but 'intermolecular attraction = 0' whereas the drawing in Figure 5(b) was coded as '1' in both. In Figure 6, the two iodine molecules, given as I-I are placed amongst the water molecules representing iodine dispersed in water, yet Figure 5(b) shows a similar scene for solution because iodine particles are surrounded by water molecules in a certain orientation. In this figure, oxygen atoms are placed facing toward iodine molecules, and hydrogen atoms are shown facing outwards forming a hydration shell depicting attraction between iodine and water molecules.

Particulate-level features that appeared more often in written responses. Features that appeared significantly more often in written responses to particulate-level items involved *dynamics*: either *processes* such as dissolving, molecular *motion*, or features involving both, such as pressure (Figure 7). Figure 7(a) shows how the students attempted to communicate in pictures or words how molecules break free of the liquid state in evaporation. Figure 7(b) shows how students described the nature of a saturated solution either pictorially or in words. In Figure 7(c), students illustrated four stages in chemical equilibrium in either a series of pictures or a list of written steps. Specifically, in Figure 7(a), the drawing (on the left) includes arrows depicting evaporation and condensation, and the written explanation (on the right) describes evaporation by stating how molecules behave, that is, *break free* from the surface. In

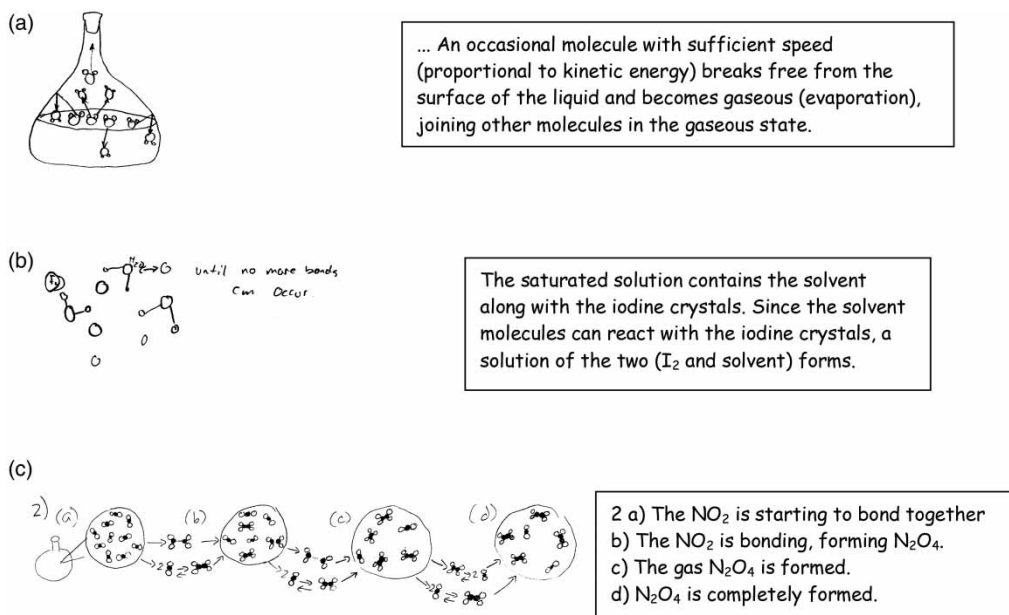


Figure 7. Sample pictorial and written responses showing *process* for the particulate-level explanations of (a) liquid–vapor equilibrium, (b) solubility equilibrium, and (c) chemical equilibrium. Notice how arrows were used in (a) and (b) to indicate motion and in (c) to indicate sequence

Figure 7(b), the pictorial response (on the left) shows the iodine and water molecules dispersed and *interacting* with each other shown by a double-sided arrow and the label, *until no more bonds can occur*. The written response (on the right) of Figure 7(b) describes the processes by stating the *reaction* taking place between the solvent molecules and iodine crystals. In Figure 7(c), the drawing (on the left) shows the processes of *formation* of N_2O_4 from the combination of two NO_2 molecules and the *decomposition* of N_2O_4 into two NO_2 molecules by double arrows. The written explanation (on the right) of Figure 7(c) includes the process of formation of N_2O_4 by describing *bonding* in words.

The more frequent appearance of process and motion features in written responses could be related to the ease of using language tools to describe more abstract concepts and dynamic relations. This finding is parallel with Tversky's (2001) suggestion that written languages are more likely to be used to describe concepts difficult to depict visually. It should be noted that restricting participants to words or pictures did not necessarily prevent them from providing complete responses. For example, notice that in Figure 5(c) both the drawing and the text convey essentially the same information, but in different ways. Nevertheless, most of the papers did present incomplete information.

Physical and structural features, such as the liquid phase, surface, hydrogen bonds, flask, and the ratio of reactant to product, appeared significantly more frequently in pictorial explanations than in written ones; whereas process and motion appeared more frequently in written explanations at the particulate level. Specifically, when the responses of instructors were compared, it was found that at the macroscopic level, physical or macroscopic structures such as liquid, surface, sealed flask, and gas were shown significantly more often in pictorial representations than in written explanations for all three equilibrium cases. On the other hand, features describing processes such as formation of hydrogen bonds and equilibrium were represented significantly more often in written explanations than in pictorial representations. Also at the particulate level, participants were more likely to show physical and structural features such as liquid phase, surface, hydrogen bond, flask, and ratio in pictorial responses than in written ones; whereas process and motion appeared more frequently in written explanations. Interviews with instructors revealed that whether the communication was written or pictorial each instructor believed that a complete answer had been provided. Table S2, showing the results of the detailed analysis of instructor responses, can be found in the supplementary materials.

The comparison of written and pictorial student responses at macroscopic and particulate levels revealed very similar findings. In addition, a number of misconceptions were noted in the student explanations. Misconceptions related to the dynamic nature of equilibrium, such as a reaction going to completion or proceeding after equilibrium, were represented significantly more often in drawings than in written explanations. Table S3, which shows the results of the detailed analysis of student responses, is included in the supplementary materials.

In the analysis of the drawings it was useful to note how pictorial elements, such as lines and arrows, spatial arrangement, and occasionally labels and written

explanations, were used to convey meaning. Lines were used to depict macroscopic features such as the shape of a container, and to indicate molecular motion, arrows were used to represent processes, sequence, labeling, and motion, and spatial arrangement was used to depict proximity and to indicate molecular motion and interactions. Some pictorial representations included text to provide labels or explanations. Finally, individual preferences in type of molecular representations were observed; for example, some participants drew space-filling models, some drew elemental symbols represented by lines, and some drew only circles.

Conclusions

Chemistry instructors, when teaching chemistry, and students when explaining the concepts, make use of both pictorial and written explanations. However, there are times when only words or pictures are used. For example, an instructor may present a lecture without pictures or respond to a student question using only words, and animations of molecular behavior may be produced without narration. In addition, research on student mental models of chemical concepts may examine only written explanations or only pictorial representations. This investigation found that pictorial and written explanations of physical and chemical equilibria can convey different meanings and may emphasize different features of the same phenomenon. Regardless of chemistry background, the medium used for representations may influence the information a person conveys.

This study revealed that features found in pictorial and written representations were different for both the macroscopic- and particulate-level explanations of equilibrium. At the macroscopic level, pictorial representations included physical or observable structures, such as the presence of a solid, significantly more often; whereas written representations for the same level included features of motion and processes such as evaporation, saturation, and equilibrium significantly more often and were more likely to refer to substances by name or formula. For both the macroscopic and particulate levels, pictorial representations included physical or external objects, such as flask or solid, for all the equilibrium conditions. Similarly, even when subjects were asked to represent the situation at the particulate level, macroscopic features such as flask, surface, and solution were shown significantly more often in the diagrams than they were expressed in written form.

At the particulate level, similar results were found: pictorial representations were more likely to portray physical or macroscopic *structures* such as the liquid phase, surface, and homogeneity (appearance), as well as particulate structures such as hydrogen bonds, formulae of molecules, and equilibrium ratio; whereas the written explanations were significantly more likely to mention features involving *dynamics* such as motion, processes, or both. Tversky (2005) argues for the superiority of image-like representations over language-like ones due to representation of conceptual and spatial relations. This argument is parallel with the findings of this study because some particulate features such as favorable orientation of molecules, which were never mentioned in written representations, were observed in drawings.

In addition, molecular structures were shown more often in the pictorial representations than they were expressed in writing. These findings are consistent with the suggestions of cognitive psychologists such as Tversky (2005) and Schnotz (2001), who claim that written descriptions are more general and abstract, whereas pictures are more concrete and specific. Our findings also showed that pictorial representations of the macroscopic level focused mostly on the visible, concrete features, while written ones conveyed more abstract and dynamic features than would be seen with the eye. It is important to note that participants, even experienced instructors, believed that they were providing a complete explanation. However, the results showed that each type of communication had a different focus and participants using each type omitted key information.

When the pictorial and written responses were compared across expertise, that is, for instructors and students of chemistry, the results were similar to what was found for the whole group: Physical structures were emphasized more often in pictorial explanations and dynamics, motion, and process more often in written representations. This finding suggests that the affordances of each type of representation may not be easily overcome by the level of chemistry knowledge. Therefore, these affordances should be considered when the medium of representations is restricted. The expressed mental models of the participants inferred from the researchers' interpretation of pictorial responses would have in most cases indicated a model showing molecules in a static situation rather than a dynamic equilibrium, but interacting with one another. The researchers interpreted the written responses to reveal an understanding of the dynamic nature of equilibrium, but at times without a clear depiction of molecular structure and interaction and with fewer connections between the macroscopic and particulate levels.

The novel contribution of this study is recognizing the extent to which the mode of communication affects the nature of the explanations people generate to express their mental models, and specifically the visual tools used, regardless of expertise in chemistry. This result highlights the importance of recognizing which cognitive tools are used in communicating ideas via representations of chemical phenomena. It also suggests that to give learners a fuller understanding of chemical phenomena both words and pictures should be used in instruction.

Implications for Teaching and Learning Chemistry

Science in general and chemistry in particular involve understanding and using multiple representations, including text, drawings, table, graphs, and symbols. Instructors and students are expected to know and use these representations, and to be able to translate from one to another. As the results of the study imply, different representations may have different affordances, and these should be considered while teaching and designing learning environments.

Taking into consideration features that are better represented by written or pictorial explanations could help students gain better conceptual understanding. For example, drawings could be used when describing structural features such as

hydrogen bonding, and animations of processes such as the increase in molecular motion with temperature could be accompanied by explanatory narration or worksheets (Akaygun & Jones, 2013a). Theory, as well as instructor intuition, suggests that these different representations should not be emphasized separately, but rather interconnected. According to the dual coding theory, when text and visual components are integrated, learning is enhanced (Paivio, 1969). Consistent with Paivio's theory, Treagust, Chittleborough, and Mamiala (2003) found the simultaneous use of different types of representation and text to be important for the learning of chemistry.

The results of this study may also have implications for assessment. Because participants were more likely to show macroscopic and structural features in their depictions and process and motion features in their written explanations, instructors may find it more effective to assess dynamic features by asking for written answers; whereas when they evaluate student understanding of structure and macroscopic properties, they may ask students to generate drawings. Instructors attempting to discover student misconceptions might also find that questions requiring both text and drawings reveal more types of misconceptions than either text or drawings alone. Instructors can also challenge learners to expand their representational skills. For instance, students may purposefully be asked to draw processes.

In their drawings, the participants in this study displayed individual preferences for different kinds of molecular representations, such as symbolic, circular, or molecular structure. This finding suggests that chemistry instructors should make use of a variety of styles of molecular representations in their teaching. Different visual media, such as pictures, drawings, videos, and computer animations, could be used for this purpose. Computer animations and simulations at the macroscopic and particulate levels have been shown to help students enhance their understanding of chemical processes (Akaygun, 2009; Ardac & Akaygun, 2004; Burke, Greenbowe, & Windschitl, 1998; Kelly & Jones, 2007; Sanger, Phelps, & Fienhold, 2000; Williamson & Abraham, 1995). Akaygun and Jones (2013b) reviewed research on the effects of dynamic visualizations used to depict structure and processes at the particulate level. Overall, the impact of using dynamic computer visualizations on students' understanding of chemical phenomena was found to be positive.

Because learning chemistry involves understanding chemical phenomena at macroscopic, symbolic, and particulate levels (Johnstone, 1993), making use of various types of representations may help learners better connect the levels to one another. For instance, to develop particulate-level understanding, students may be asked to draw molecular structures but to explain the processes in words. Written and pictorial representations are both effective tools for conceptualizing chemistry when instructors know how best to use each type of tool.

Implications for Research

The ability to characterize the mental models of participants by means of their drawings or written explanations may help researchers understand how people visualize

phenomena at different levels. Understanding how people represent the different features of a concept or system may help researchers know when it is better to request drawings and when it is better to request written explanations. The results of this study suggest that the use of either drawings or written explanations alone to assess understanding of the dynamic nature of equilibrium may not reveal critical aspects of mental models. In future research, mental models of experts and novices will be compared in detail in order to develop and appropriately evaluate more effective instructional materials. In addition, this study could be repeated for different areas and levels of chemistry in order to reach a more comprehensive understanding of how people make use of different types of representations to describe different phenomena.

Implications for Developers and Designers

The findings of this study suggest that both written and pictorial representations need to be considered when designing and developing dynamic computer visualizations. In a study by Mayer and Anderson (1992), students benefitted the most when pictures and words were presented contiguously, as suggested by the dual coding theory. Because this study found that process features were represented more frequently in written form, instructional designers could accompany dynamic visualizations with text or narration that specifically focuses attention on process features. They can also address different learning preferences by using a variety of points of view and enhance clarity by being explicit about the functions of different types of arrows.

Acknowledgements

The authors would like to acknowledge support from the National Science Foundation (NSF) award number REC-0440103, Design Principles for Effective Molecular Animations. They would also like to thank Barbara Tversky for taking the time to introduce them to cognitive science research strategies and for her encouragement and many helpful suggestions. Gratitude is also due to Roy Tasker, Jerry Honts, Jerry Suits, David Falvo, David Pringle, Robin Macaluso, Heng-Yu Ku, Dilek Ardac, and the University of Northern Colorado Chemical Education Research Group for their willingness to review and comment on all aspects of this study.

References

- Ainsworth, S. (1999). The functions of multiple representations. *Computers & Education*, 33, 131–152.
- Akaygun, S. (2009). *The effect of computer visualizations on students' mental models of the dynamic nature of physical equilibrium* (Unpublished doctoral dissertation). Greeley, CO: University of Northern Colorado.
- Akaygun, S., & Jones, L. L. (2013a). How does level of guidance affect understanding when students use a dynamic simulation of liquid–vapor equilibrium? In I. Devetak, S. A. Glazar, & L. Plut-Pregelj (Eds.), *Active learning and understanding in the chemistry classroom*. Springer.

- Akaygun, S., & Jones, L. L. (2013b). Dynamic visualizations: Tools for understanding the particulate nature of matter. In G. Tsapalis & H. Sevian (Eds.), *Concepts of matter in science education* (pp. 282–300). Dordrecht, The Netherlands: Springer.
- Ardac, D., & Akaygun, S. (2004). Effectiveness of multimedia-based instruction that emphasizes molecular representations on students' understanding of chemical change. *Journal of Research in Science Teaching*, 40(4), 317–337.
- Ardac, D., & Akaygun, S. (2005). Using static and dynamic visuals to represent chemical change at molecular level. *International Journal of Science Education*, 27(11), 1269–1298.
- Banerjee, A. (1995). Teaching chemical equilibrium and thermodynamics in undergraduate general chemistry classes. *Journal of Chemical Education*, 72(10), 879–881.
- Ben-Zvi, R., Eylon, B., & Silberstein, J. (1986). Is an atom of copper malleable? *Journal of Chemical Education*, 63(1), 64–66.
- Bergquist, W., & Heikkinen, H. (1990). Student ideas regarding chemical equilibrium. *Journal of Chemical Education*, 67(12), 1000–1003.
- Bodner, G. M., Gardner, D. E., & Briggs, M. W. (2005). Models and modeling. In N. Pienta, M. Cooper, & T. Greenbowe (Eds.), *Chemists' guide to effective teaching* (pp. 67–76). Upper Saddle River, NJ: Prentice-Hall.
- Burke, K., Greenbowe, T., & Windschitl, M. (1998). Developing and using conceptual computer animations for chemistry instruction. *Journal of Chemical Education*, 75(12), 1658–1661.
- Camacho, M., & Good, R. (1989). Problem solving and chemical equilibrium: Successful versus unsuccessful performance. *Journal of Research in Science Teaching*, 26(3), 251–272.
- Cheng, M., & Gilbert, J. K. (2009). Towards a better utilization of diagrams in research into the use of representative levels in chemical education. In J. K. Gilbert & D. Treagust (Eds.), *Multiple representations in chemical education, Models and modeling in science education* (pp. 55–73). Heidelberg: Springer.
- Chiu, M.-H., Chou, C.-C., & Liu, C.-J. (2002). Dynamic processes of conceptual change: Analysis of constructing mental models of chemical equilibrium. *Journal of Research in Science Teaching*, 39(8), 688–712.
- Coll, R. K., & Treagust, D. F. (2003). Investigation of secondary school, undergraduate, and graduate learners' mental models of ionic bonding. *Journal of Research in Science Teaching*, 40(5), 464–486.
- Craik, K. (1943). *The nature of explanation*. Cambridge: Cambridge University Press.
- Davidowitz, B., Chittleborough, C., & Murray, E. (2010). Student generated submicro diagrams: A useful tool for teaching and learning chemical equations and stoichiometry. *Chemical Education Research and Practice*, 11, 154–164.
- Ebenezer, J., & Gaskell, J. (1995). Relational conceptual change in solution chemistry. *Science Education*, 79(1), 1–17.
- Ehrlén, K. (2009). Drawings as representations of children's conceptions. *International Journal of Science Education*, 31(1), 41–57.
- Gilbert, J. K. (1997). *Exploring models and modeling in science education and technology education*. Reading, UK: The University of Reading.
- Gilbert, J. K., & Treagust, D. (2009). Introduction: Macro, submicro and symbolic representations and the relationship between them: Key models in chemical education. In J. K. Gilbert & D. Treagust (Eds.), *Multiple representations in chemical education* (pp. 1–8). Dordrecht, The Netherlands: Springer.
- Gobert, J. D. (2005). Leveraging technology and cognitive theory on visualization to promote students' science. In J. K. Gilbert (Ed.), *Visualization in science education* (pp. 73–90). Dordrecht, The Netherlands: Springer.
- Greca, I. M., & Moreira, M. A. (2000). Mental models, conceptual models, and modeling. *International Journal of Science Education*, 22, 1–11.
- Gussarsky, E., & Gorodetsky, M. (1988). On the chemical equilibrium concept: Constrained word associations and conception. *Journal of Research in Science Teaching*, 25, 319–333.

- Gussarsky, E., & Gorodetsky, M. (1990). On the concept "chemical equilibrium": The associative framework. *Journal of Research in Science Teaching*, 27, 197–204.
- Hackling, M. W., & Garnett, P. J. (1985). Misconceptions of chemical equilibrium. *European Journal of Science Education*, 7(2), 205–214.
- Haidar, A. H., & Abraham, M. R. (1991). A comparison of applied and theoretical knowledge of concepts based on the particulate nature of matter. *Journal of Research in Science Teaching*, 28(10), 919–938.
- Harrison, A. G., & Treagust, D. F. (2002). The particulate nature of matter: Challenges in understanding the submicroscopic world. In J. K. Gilbert (Ed.), *Chemical education: Toward research-based practice* (pp. 189–212). Dordrecht, The Netherlands: Kluwer Academic Publishers.
- Ingham, A. M., & Gilbert, J. K. (1991). The use of analogue models by students of chemistry at higher education level. *International Journal of Science Education*, 13(2), 193–202.
- Johnstone, A. H. (1993). The development of chemistry teaching: A changing response to changing demand. *Journal of Chemical Education*, 70(9), 701–704.
- Jones, L. L., Jordan, K. D., & Stillings, N. A. (2005). Molecular visualization in chemistry education: The role of multidisciplinary collaboration. *Chemistry Education Research and Practice*, 6(3), 136–149.
- Justi, R., & Gilbert, J. K. (2002). Models and modelling in chemical education. In J. K. Gilbert (Ed.), *Chemical education: Toward research-based practice* (pp. 47–68). Dordrecht, The Netherlands: Kluwer Academic Publishers.
- Kelly, R. M., & Jones, L. L. (2007). Exploring how different features of animations of sodium chloride dissolving affect students' explanations. *Journal of Science Education and Technology*, 16(5), 413–429.
- Kelly, R. M., & Jones, L. L. (2008). Investigating students' ability to transfer ideas learned from molecular animations of the dissolution process. *Journal of Chemical Education*, 85, 303–309.
- Kern, A. L., Wood, N. B., Roehrig, G. H., & Nyachwaya, J. M. (2010). A qualitative report of the ways high school chemistry students attempt to represent a chemical reaction at the atomic/molecular level. *Chemical Education Research and Practice*, 11, 154–164.
- Kozma, R. B., & Russell, J. (1997). Multimedia and understanding: Expert and novice responses to different representations of chemical phenomena. *Journal of Research in Science Teaching*, 34(9), 949–968.
- Lekhavat, P., & Jones, L. (2009). The effect of adjunct questions emphasizing the particulate nature of matter on students' understanding of chemical concepts in multimedia lessons. *Educación Química*, 20(3), 351–359.
- Lesh, R., Hover, M., Hole, B., Kelly, A., & Post, T. (2000). Principles for developing thought-revealing activities for students and teachers. In A. Kelly & R. Lesh (Eds.), *Handbook of research design in mathematics and science education* (pp. 591–645). Mahwah, NH: Lawrence Erlbaum.
- Margel, H., Eylon, B. S., & Scherz, Z. (2008). A longitudinal study of junior high school students' conceptions of the structure of materials. *Journal of Research in Science Teaching*, 45(1), 132–152.
- Mayer, R. E. (2001). *Multimedia learning*. Cambridge: Cambridge University Press.
- Mayer, R. E., & Anderson, R. B. (1992). The instructive animation: Helping students build connections between words and pictures in multimedia learning. *Journal of Educational Psychology*, 84(4), 444–452.
- Nakhleh, M. B. (1992). Why some students don't learn chemistry. *Journal of Chemical Education*, 69(3), 19–196.
- Niaz, M. (1995). Relationship between student performance on conceptual and computational problems of chemical equilibrium. *International Journal of Science Education*, 17, 343–355.
- Noh, T., & Scharmann, L. C. (1997). Instructional influence of a molecular-level pictorial presentation of matter on students' conceptions and problem-solving ability. *Journal of Research in Science Teaching*, 34(2), 199–217.

- Nurrenbern, S. C., & Pickering, M. (1987). Concept learning versus problem solving: Is there a difference? *Journal of Chemical Education*, 64(6), 508–509.
- Nyachwaya, J. M., Mohamed, A.-R., Roehrig, G. H., Wood, N. B., Kern, A. L., & Schneider, J. L. (2011). The development of an open-ended drawing tool: An alternative diagnostic tool for assessing students' understanding of the particulate nature of matter. *Chemistry Education Research and Practice*, 12, 121–132.
- Osborne, R. J., & Cosgrove, M. M. (1983). Children's conceptions of the changes of state of water. *Journal of Research in Science Teaching*, 20(9), 825–838.
- Paivio, A. (1969). Mental imagery in associative learning and memory. *Psychological Review*, 76(3), 241–263.
- Paivio, A. (1990). *Mental representations: A dual coding approach*. New York: Oxford Science Publications.
- Prain, V., & Tytler, R. (2012). Learning through constructing representations in science: A framework of representational construction affordances. *International Journal of Science Education*, 34(17), 2751–2773.
- Sanger, M. (2000). Using particulate drawings to determine and improve students' conceptions of pure substances and mixtures. *Journal of Chemical Education*, 77(6), 762–766.
- Sanger, M., Phelps, A., & Fienhold, J. (2000). Using a computer animation to improve students' conceptual understanding of a can-crushing demonstration. *Journal of Chemical Education*, 77(11), 1517–1520.
- Schnotz, W. (2001). Sign system, technologies, and the acquisition of knowledge. In J. F. Rouet, J. Levonen, & A. Biarreau (Eds.), *Multimedia learning* (pp. 9–29). Amsterdam: Elsevier Science.
- Smith, K. J., & Metz, P. A. (1996). Evaluating student understanding of solution chemistry through microscopic representations. *Journal of Chemical Education*, 73(3), 233–237.
- Treagust, D. F., Chittleborough, G., & Mamiala, T. L. (2003). The role of submicroscopic and symbolic representations in chemical explanations. *International Journal of Science Education*, 25, 1353–1368.
- Tversky, B. (2001). Spatial schemas in depictions. In P. Grialou, G. Longo, & M. Okado (Eds.), *Spatial schemas and abstract thought* (pp. 79–112). Cambridge: MIT Press.
- Tversky, B. (2005). Some ways images express and promote thought. In P. Grialou, G. Longo, & M. Okado (Eds.), *Image and reasoning* (pp. 15–29). Tokyo: Keiko University Press.
- Tversky, B., Agrawala, M., Heiser, J., Lee, P. U., Hanrahan, P., Phan, D., . . . Daniele, M. (2006). Cognitive design principles for automated generation of visualizations. In G. Allen (Ed.), *Applied spatial cognition: From research to cognitive technology* (pp. 53–73). Mahwah, NJ: Erlbaum.
- Tyson, L., Treagust, D. F., & Bucat, R. B. (1999). The complexity of teaching and learning chemical equilibrium. *Journal of Chemical Education*, 76(4), 554–558.
- Van Driel, J. H., & Gräber, W. (2002). The teaching and learning of chemical equilibrium. In J. K. Gilbert, O. De Jong, R. Justi, D. F. Treagust, & J. H. Van Driel (Eds.), *Chemical education: Towards research-based practice* (pp. 271–292). Dordrecht, The Netherlands: Kluwer Academic Publishers.
- Voska, K. W., & Heikkinen, H. W. (2000). Identification and analysis of student conceptions used to solve chemical equilibrium problems. *Journal of Research in Science Teaching*, 37(2), 160–176.
- Wheeler, A. E., & Kass, H. (1978). Student misconceptions in chemical equilibrium. *Science Education*, 62(2), 223–232.
- Williamson, V. (2008). The particulate nature of matter: An example of how theory-based research can impact the field. In D. Bunce & R. S. Cole (Eds.), *Nuts and bolts of chemical education research* (pp. 67–78). Washington, DC: American Chemical Society.
- Williamson, V. M., & Abraham, M. R. (1995). The effects of computer animation on the particulate mental models of college chemistry students. *Journal of Research in Science Teaching*, 32(5), 521–534.